Analysis and Classification of Dysarthric Speech

Vartika Tyagi
IT dept.
Indira Gandhi Delhi Technical University for Women
Delhi, India
vartikatyagi400@gmail.com

Amita Dev
IT dept.
Indira Gandhi Delhi Technical University for Women
Delhi, India
vc@igdtuw.ac.in

Poonam Bansal
AI & DS dept.
Indira Gandhi Delhi Technical University for Women
Delhi, India
poonambansal@igdtuw.ac.in

2023 26th Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA) | 979-8-3503-4402-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/O-COCOSDA60357.2023.10482956

***Abstract*—Classifying dysarthria using neural networks is challenging due to several factors inherent to the nature of dysarthria, the complexity of speech signals, and the requirements of effective machine learning models. Impaired speech classification is challenging for two main reasons: firstly, the data is scarce, and secondly, it is heterogeneous. In this paper, we have trained two different architectures on a dysarthric speech database. A comparison of results shows that according to precision, recall, f1-score, and accuracy, the Deep Neural Network (DNN) model outperforms the classical Convolutional Neural Network (CNN) model, even with a small database. For people with dysarthria, a DNN can improve the performance metrics during the classification of impaired speech by 99% compared to the classical architecture. This improvement is more than that provided by CNN. We have used the TORGO database for this work.**

Keywords—Deep Neural Network, Convolutional Neural Network, speech classification, dysarthria, impaired speech

I. INTRODUCTION

In our day-to-day life, social interaction ought to be the most essential [17]. People often anticipate speaking with machines to convert spoken utterances into text [18]. Classifying dysarthria using neural networks involves training a model to automatically recognize and classify different types or severity levels based on input features, like acoustic features which are extracted from linguistic features, speech signals, or a combination of both. A variety of neurocognitive problems resulting from traumatic brain injury (TBI), cerebral palsy, stroke, amyotrophic lateral sclerosis, dementia, myasthenia gravis, and Parkinson's disease can lead to dysarthria, a kind of speech impairment [4]. Impaired speech classification is important because it encourages early diagnosis, personalized therapy, analysis, and the expansion of assistive technologies, all of which enhance the quality of life and inclusion of individuals with speech disorders. It also plays a role in legal and educational support and discloses healthcare findings associated with speech impairments. In this paper, we have classified dysarthric audio using CNN and DNN. This document is formatted as follows: Details on the database we utilize for our work are given in Section II. Next, Section III explains the fundamental structure of a classification process and the particular methods and techniques we employ. Section IV highlights the CNN and DNN models and displays the various classification tasks we have performed. Section V contains the results and discussion, while Section VI offers a conclusion and recommendations for more study.

II. DATASET

The normal utterance of sounds can be affected by many speech disorders. Dysarthric speakers exhibit problems articulating phonemes due to damage to the nervous system. This may result in altered vocal quality, sluggish speech tempo, or irregular pitch and rhythm. The University of Toronto's TORGO dysarthric database contains data from eight dysarthric subjects (five men and three women) who have speech impairment brought on by amyotrophic lateral sclerosis or cerebral palsy, approximately 13.5 hours of audio data which has 16394 utterances, as well as data from seven control speakers (four men and three women). It is made up of 7.7 hours of single-word utterances and 5.8 hours of brief sentences.

III. DATA VISUALIZATION AND FEATURE EXTRACTION

A. Data Visualization and EDA

1) Wave plots: A wave plot, in the context of audio and signal processing, is a graphical representation of a waveform, which is a visualization of a one-dimensional signal over time. These are commonly used to depict audio signals, including speech and music, and they provide insights into the signal's amplitude (strength) characteristics as it varies over time.

For the dysarthric samples, it can be observed that the one male sample taken here has a slurred speech, as is evident by the wave plot shown in Fig. 1, however, because the words in the female sample overlap, her speech is quite fast and challenging to comprehend with each other as is indicative by the box like patterns forming in the wave plot in Fig. 2. A waveplot, in the context of audio and signal processing, is a graphical representation of a waveform, which is a visualization of a one-dimensional signal over time. These are commonly used to depict audio signals, including speech and music, and they provide insights into the characteristics of the signal's amplitude (strength) as it varies over time.

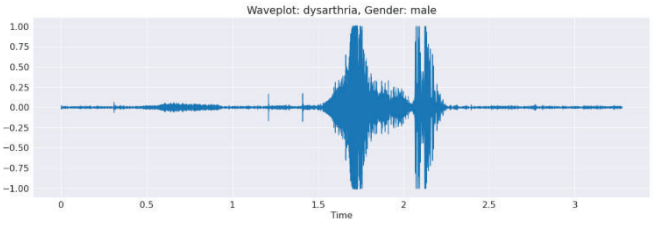
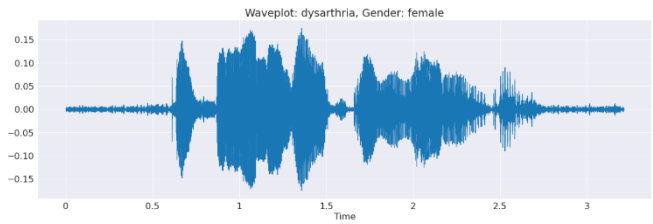


Fig. 1. Wave plot of dysarthric male audio



ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:37:16 UTC from IEEE Xplore. Restriction

Fig. 2. Wave plot of dysarthric female audio

The non-dysarthric samples have a regular wave plot indicating a regular-paced speech, as shown in Fig. 3 and Fig. 4 mentioned below:

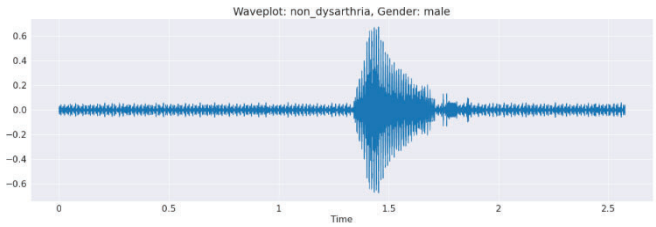


Fig. 3. Wave plot of non-dysarthric male audio

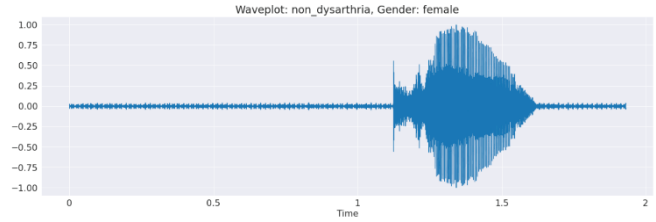


Fig. 4. Wave plot of non-dysarthric female audio

2) Spectrogram: It provides a visual depiction of how a signal's range of frequencies changes over time. Spectrograms are widely used in audio and signal processing to analyze and visualize the frequency content of time-varying signals, such as audio recordings or other time-domain data.

It is clear that the energy magnitudes of the frequencies for the dysarthric samples are more dispersed throughout time, which may be a sign of slurred or delayed speech or of syllables being said more quickly and overlapped. Comparable patterns (Figs. 5 and 6) might also occur in a monotonous speaker with dysarthria.

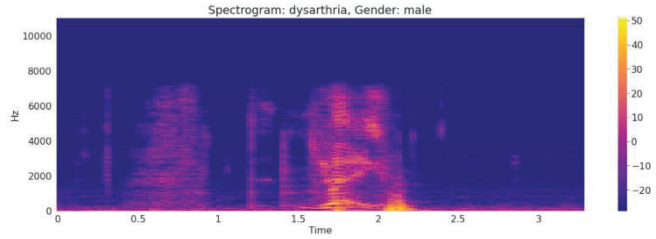


Fig. 5. Spectrogram of dysarthric male audio

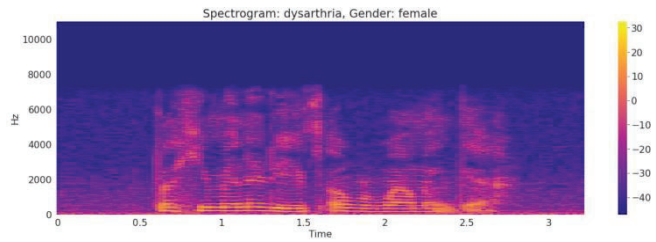


Fig. 6. Spectrogram of dysarthric female audio

Figures 7 and 8 show that the energy magnitudes for the non-dysarthric samples are limited to the portions where they have spoken, and this is done in a regular paced manner.

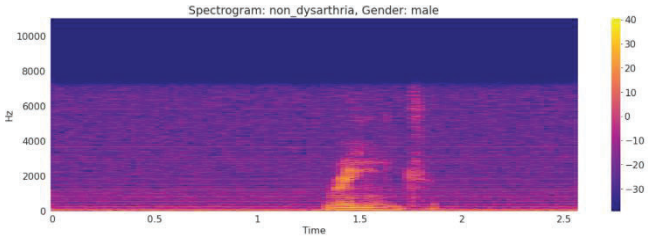


Fig. 7. Spectrogram of non-dysarthric male audio

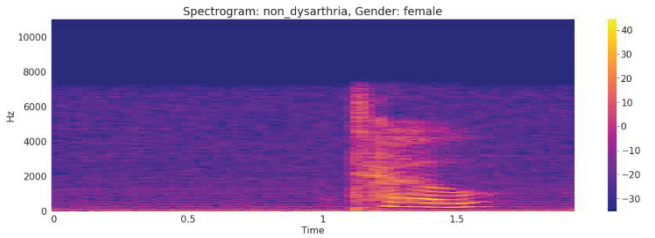


Fig. 8. Spectrogram of non-dysarthric female audio

3) Spectral Centroid: The average frequency of a sound signal is a term frequently used in audio signal processing and music analysis. It tells us where a sound's "brightness" or amount of energy is focused in its frequency range. Spectral Centroid is often used as a feature for various audio classification and analysis tasks. The centroids are usually high when there is no speech and become low when speech is present in the audio.

For dysarthric samples, it can be seen that since the speech is slurred, the centroid values go down for a while and go back up as soon as there are some small pauses, as demonstrated in Fig. 9 and Fig. 10:

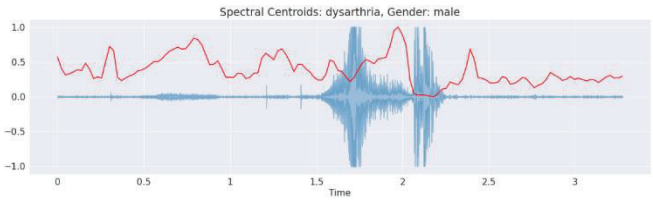


Fig. 9. Spectral Centroids of dysarthric male audio

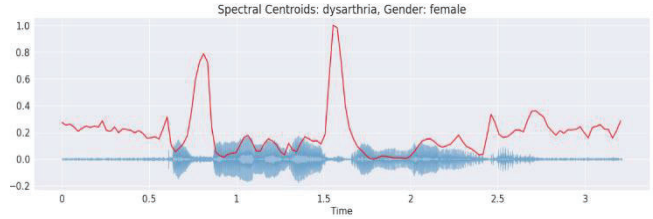


Fig. 10. Spectral Centroids of dysarthric female audio

For non-dysarthric samples, the centroids are down for a little while where the speech is present in the audio, and they go back up again as soon as there is no speech in the audio, as mentioned in Fig. 11 and Fig. 12 below:

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:37:16 UTC from IEEE Xplore. Restriction

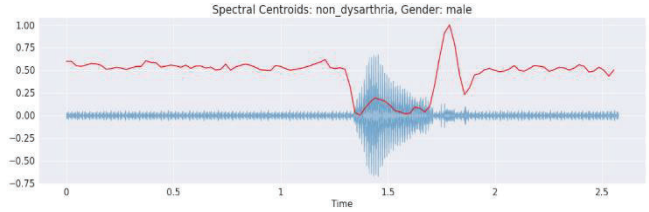


Fig. 11. Spectral Centroids of non-dysarthric male audio

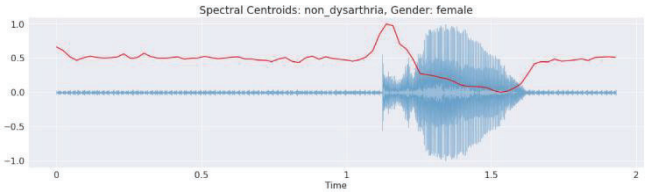


Fig. 12. Spectral Centroids of non-dysarthric female audio

4) Spectral Rolloff: It is a feature that is widely applied in music analysis and audio signal processing to identify the frequency below which a particular percentage of the signal's total spectral energy is present. It measures the spectral shape or distribution of power in an audio signal and provides information about the "roll-off" point in the frequency spectrum. Spectral Rolloffs show the frequency below which a portion of the magnitude distribution is concentrated.

As we had previously discovered in the spectrograms, the energy magnitudes of the frequencies are more distributed in the case of dysarthric samples; in Figs. 13 and 14, the speech contained in the audio signal is surrounded by an 85% coverage below the rolloff frequency.

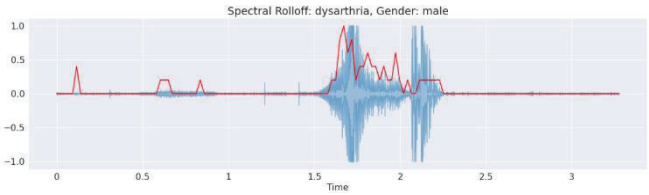


Fig. 13. Spectral Rolloff of dysarthric male audio

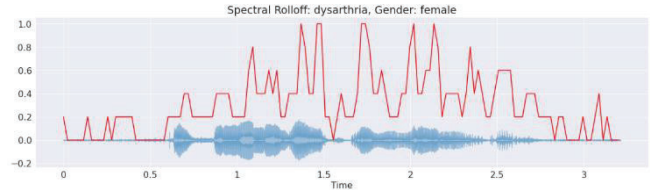


Fig. 14. Spectral Rolloff of dysarthric female audio

The speech and some background noise/disturbances in the audio are enclosed by the 85% coverage, as can be shown in Figs. 15 and 16 for non-dysarthric samples.

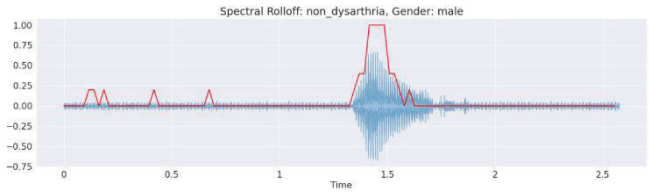


Fig. 15. Spectral Rolloff of non-dysarthric male audio

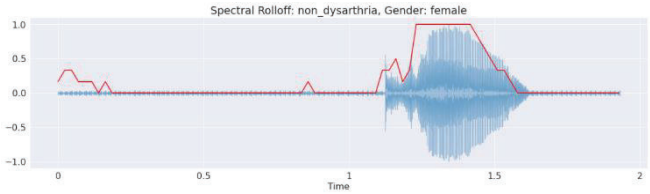


Fig. 16. Spectral Rolloff of non-dysarthric female audio

5) MFCC: MFCC stands for Mel-Frequency Cepstral Coefficients. On the mel scale, the frequency bands are evenly spaced, simulating the response of the human auditory system [14] . They are widely used in music information retrieval (MIR) and automated speech recognition (ASR) because they are good at capturing the spectral properties of audio signals and can represent audio in a concise and discriminative way.

For dysarthric samples, the MFCC visualization has much darker and more concentrated pixels indicating the slurs present in their speech, shown in Fig. 17 and Fig. 18.

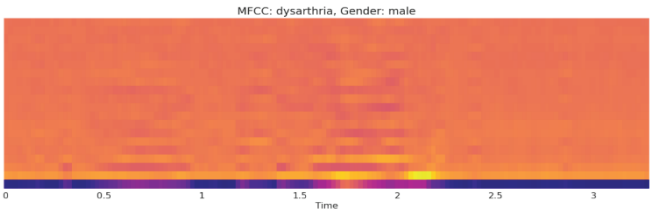


Fig. 17. MFCC for dysarthric male audio

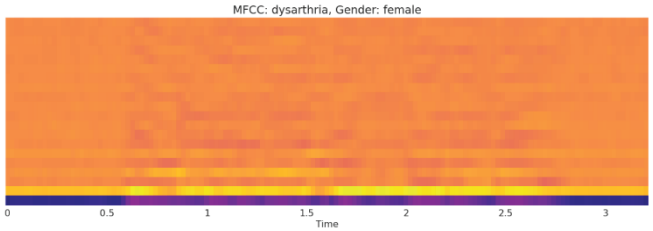


Fig. 18. MFCC for dysarthric female audio

It is not as concentrated for non-dysarthric and is more spread out, as shown in Fig. 19 and Fig. 20.

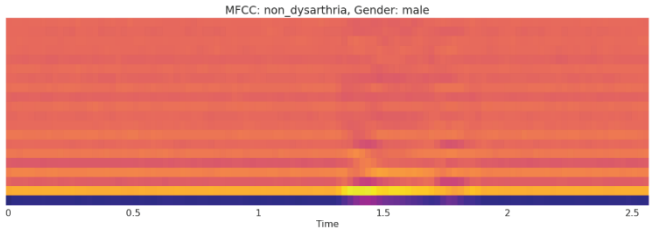


Fig. 19. MFCC for non-dysarthric male audio

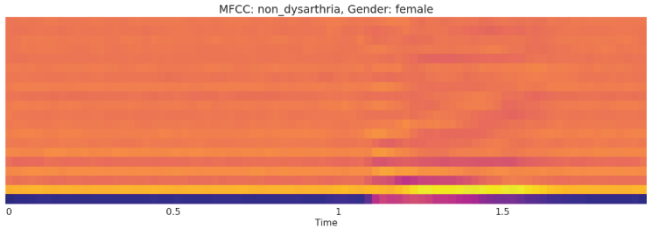


Fig. 20. MFCC for non-dysarthric female audio

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:37:16 UTC from IEEE Xplore. Restriction

6) Mel Spectrogram: The frequency content of an audio source is represented visually in a Mel Spectrogram by transforming the frequency axis to match the mel scale. It is a variation of the traditional spectrogram used in audio signal processing and analysis. The Mel spectrogram helps capture the perceptual characteristics of sound and is commonly used in speech and audio processing tasks.
The Spectrogram converted to Mel scale shows the magnitudes that are more perceptual to the human ears generally. Patterns in the mel spectrograms of the dysarthric samples indicate the slurs in the speech, as shown in Fig. 21 to Fig. 24.

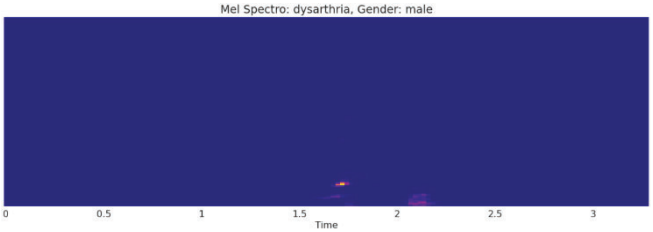


Fig. 21. Mel Spectro for dysarthric male audio

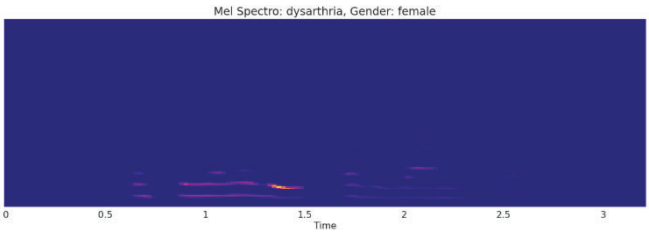


Fig. 22. Mel Spectro for dysarthric female audio

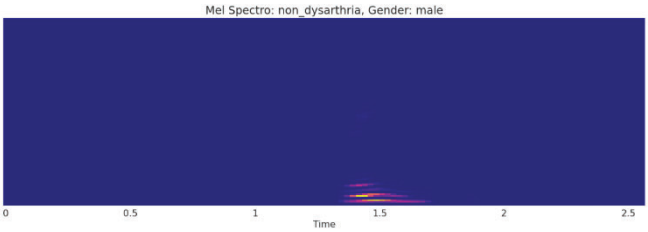


Fig. 23. Mel Spectro for non-dysarthric male audio

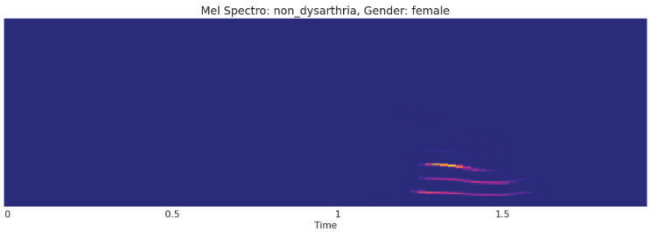


Fig. 24. Mel Spectro for non-dysarthric female audio

B. Feature Extraction

128 MFCC Features are the most features that might possibly be extracted by the authors during the process.



Fig. 29. Feature Extraction and further processing

IV. DATA MODELING

A. CNN Modeling

In order to train Convolutional Neural Networks (CNNs) to retrieve high level information, spectrograms are implemented [2] . It extracts hierarchical information from pictures using pooling and convolutional layers, which makes it very useful for image segmentation, object recognition, and classification.

B. DNN Modeling

DNN (Deep Neural Network) refers to deep neural networks with multiple hidden layers. They are versatile and used for various machine learning tasks, including structured data analysis, natural language processing, and recommendation systems. DNNs consist of densely connected layers that can capture complex data representations.

In both the models, seven neural layers are considered; below is the model summary (Fig. 25) having data that we have considered for both the models:

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 16, 8, 32)	320
max_pooling2d (MaxPooling2D)	(None, 8, 4, 32)	0
conv2d_1 (Conv2D)	(None, 8, 4, 64)	18496
max_pooling2d_1 (MaxPooling2	(None, 4, 2, 64)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 32)	16416
dense_1 (Dense)	(None, 1)	33

Total params: 35,265
Trainable params: 35,265
Non-trainable params: 0

Fig. 25. Model Summary of CNN and DNN models

Learning Curves

Following are the learning curves of model loss and model accuracy of both models:

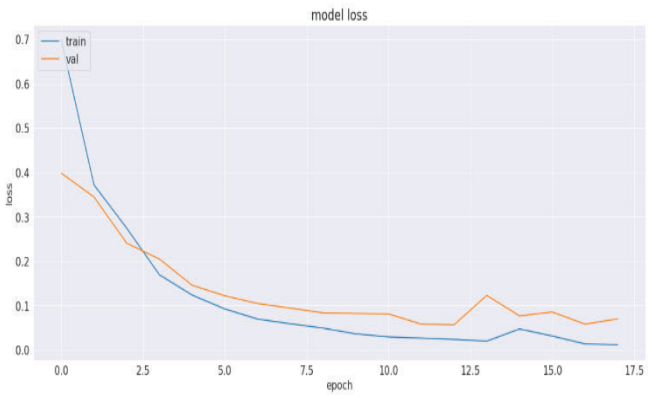


Fig. 26. CNN model loss curves

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:37:16 UTC from IEEE Xplore. Restriction



Fig. 27. CNN model accuracy curves

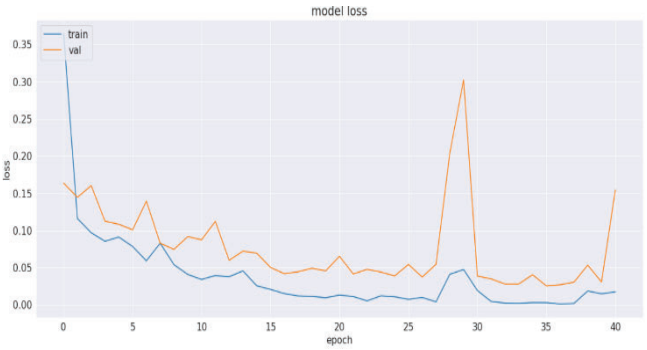


Fig. 28. DNN model loss curves



Fig. 29. DNN model accuracy curves

ROC Curve and AUC Score

AUC provides an aggregate evaluation of performance across all possible classification thresholds. The current model gives us an AUC score of 99% for CNN and 98% for DNN. Since it is close to 1, our models have excellent separability for dysarthric and non-dysarthric cases.

The ROC curve also indicates that this model will perform well with the threshold of 0.5.

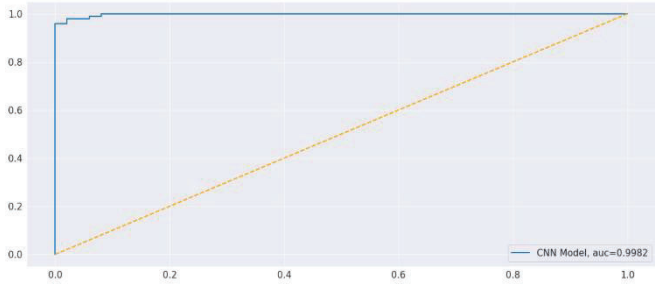


Fig. 30. CNN ROC curve and AUC score

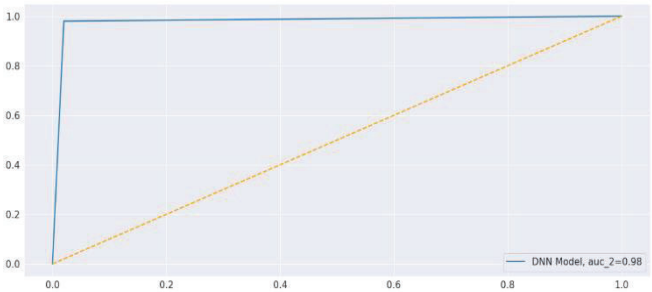


Fig. 31. DNN ROC curve and AUC score

V. Results and Discussion

We use the TORGO dataset for the dysarthria classification using CNN and DNN. If we discuss the confusion matrix of both the models, in the CNN model, 2 misclassification from each class is observed, and in DNN, only 1 misclassification from each class is attended. Upon running the models several times, the performance remains the same.

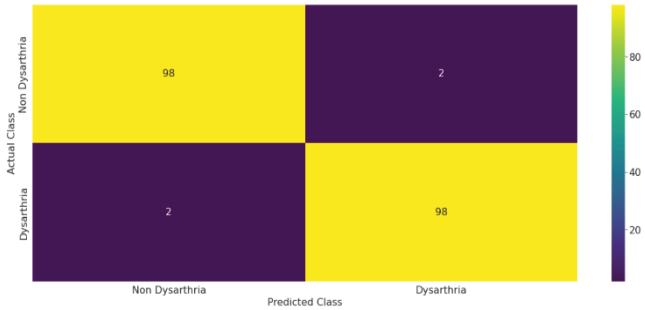


Fig. 32. Confusion matrix of CNN

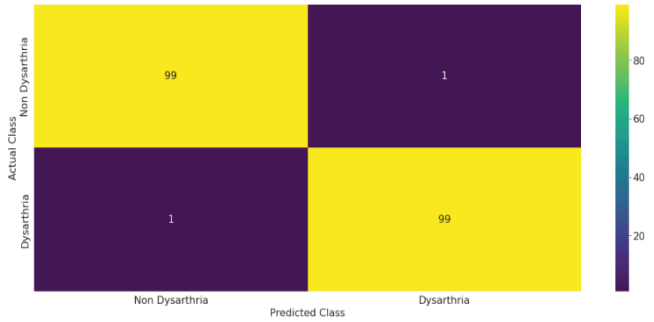


Fig. 33. Confusion matrix of DNN

Classification Report

Based on the accuracy, precision, recall, F-1, and accuracy scores for both models, we have classified the speech impairment as indicated below, and we can say that on comparing both the reports, it is clear that DNN is performing better than CNN in all aspects.

TABLE I

For CNN Model	**Precision score**	**Recall score**	**F1-Score**
0.0	0.98	0.98	0.98
1.0	0.98	0.98	0.98
Accuracy score			0.98
macro avg score	0.98	0.98	0.98
weighted avg score	0.98	0.98	0.98

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:37:16 UTC from IEEE Xplore. Restriction

TABLE II

For DNN Model	**Precision score**	**Recall score**	**F1-Score**
0.0	0.99	0.99	0.99
1.0	0.99	0.99	0.99
Accuracy score			0.99
macro avg score	0.99	0.99	0.99
weighted avg score	0.99	0.99	0.99

VI. CONCLUSION AND FUTURE WORK

The classification of dysarthric speech has been proposed in this study as a means of expanding the utilization of deep learning architectures for speech impairment categorization. The CNN and DNN architectures have both been used and compared. This extensive analysis used the TORGO dataset, which is openly accessible. Using the reference of the classification report, we can conclude that the accuracy for CNN is 98% while that of DNN is 99%. The recall score for CNN is 98%, and for DNN, it is 99%. These findings demonstrate the DNN architecture's capacity to manage the most severe dysarthria instances.

In future work, we will implement an overenthusiastic approach to try and make this score 100% to avoid any grave misclassifications we plan to implement. In addition, further training on impairment-specific tasks will be performed using a data augmentation strategy.

REFERENCES

[1] Geng, M., Xie, X., Ye, Z., Wang, T., Li, G., Hu, S., ... & Meng, H. (2022). Speaker adaptation using spectro-temporal deep features for dysarthric and elderly speech recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *30*, 2597-2611.

[2] Chandrakala, S. (2021). Investigation of DNN-HMM and Lattice Free Maximum Mutual Information Approaches for Impaired Speech Recognition. *IEEE Access*, *9*, 168840-168849.

[3] Liu, S., Geng, M., Hu, S., Xie, X., Cui, M., Yu, J., ... & Meng, H. (2021). Recent progress in the CUHK dysarthric speech recognition system. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *29*, 2267-2281.

[4] Chandrakala, S., Malini, S., & Veni, S. V. (2021). Histogram of states based assistive system for speech impairment due to neurological disorders. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *29*, 2425-243.

[5] Celin, T. M., Rachel, G. A., Nagarajan, T., & Vijayalakshmi, P. (2018). A weighted speaker-specific confusion transducer-based augmentative and alternative speech communication aid for dysarthric speakers. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *27*(2), 187-197.

[6] Celin, T. M., Nagarajan, T., & Vijayalakshmi, P. (2020). Data augmentation using virtual microphone array synthesis and multi-resolution feature extraction for isolated word dysarthric speech recognition. *IEEE Journal of Selected Topics in Signal Processing*, *14*(2), 346-354.

[7] Hu, S., Xie, X., Liu, S., Yu, J., Ye, Z., Geng, M., ... & Meng, H. (2021). Bayesian learning of LF-MMI trained time delay neural networks for speech recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *29*, 1514-1529.

[8] Hu, S., Xie, X., Cui, M., Deng, J., Liu, S., Yu, J., ... & Meng, H. (2022). Neural architecture search for LF-MMI trained time delay neural networks. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *30*, 1093-1107.

[9] Ye, Z., Hu, S., Li, J., Xie, X., Geng, M., Yu, J., ... & Meng, H. (2021, June). Development of the cuhk elderly speech recognition system for neurocognitive disorder detection using the dementiabank corpus. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 6433-6437). IEEE.

[10] Yue, Z., Loweimi, E., Christensen, H., Barker, J., & Cvetkovic, Z. (2022). Acoustic Modelling From Raw Source and Filter Components for Dysarthric Speech Recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *30*, 2968-2980.

[11] Almadhor, A., Irfan, R., Gao, J., Saleem, N., Rauf, H. T., & Kadry, S. (2023). E2E-DASR: End-to-end deep learning-based dysarthric automatic speech recognition. *Expert Systems with Applications*, *222*, 119797.

[12] Shahamiri, S. R. (2021). Speech vision: An end-to-end deep learning-based dysarthric automatic speech recognition system. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *29*, 852-861.

[13] Gurugubelli, K., Vuppala, A. K., Narendra, N. P., & Alku, P. (2020). Duration of the rhotic approximant/ɹ/in spastic dysarthria of different severity levels. *Speech Communication*, *125*, 61-68.

[14] Zaidi, B. F., Selouani, S. A., Boudraa, M., & Sidi Yakoub, M. (2021). Deep neural network architectures for dysarthric speech analysis and recognition. *Neural Computing and Applications*, *33*, 9089-9108.

[15] Kodrasi, I., & Bourlard, H. (2020). Spectro-temporal sparsity characterization for dysarthric speech detection. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, *28*, 1210-1222.

[16] Millet, J., & Zeghidour, N. (2019, May). Learning to detect dysarthria from raw speech. In *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 5831-5835). IEEE.

[17] Gambhir, P., Dev, A., Bansal, P., & Sharma, D. K. (2023). End-to-end multi-modal low-resourced speech keywords recognition using sequential Conv2D nets. *ACM Transactions on Asian and Low-Resource Language Information Processing*.

[18] BHATT, S., JAIN, A., & Dev, A. (2021). Monophone-based connected word Hindi speech recognition improvement. *Sādhanā*, *46*(2), 99.

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:37:16 UTC from IEEE Xplore. Restriction